

NANDHA KUMAR

Software Engineer | Open to Relocate

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SUMMARY

Product-focused Software Engineer with enterprise experience across Healthcare, Finance, and E-commerce. Deep expertise in scalable backend applications, API design, cloud architecture (AWS), and pragmatic platform integrations. Experienced modernizing legacy systems, building developer-friendly platform features, automating developer workflows, and shipping resilient Python cloud services that meet SLAs. Strong communicator who writes maintainable code, mentors peers, and turns ambiguous product requirements into production software.

PROFESSIONAL EXPERIENCE

Client: Centene Corporation

Backend Software Engineer (Cloud Application)

Sept 2024 - Present | Saint Louis, Missouri

- Led design and implementation of modular FastAPI microservices for membership lookup and enrichment, shared across care coordination workflows; authored OpenAPI specs, rate limiting, and API versioning to support multi-team integrations.
- Implemented OAuth2/JWT authentication with role-based policies and audit logging to meet security and compliance standards.
- Designed schema-validated payload handling with graceful degradation (retry/backoff, circuit breaker), reducing downstream error propagation by ~30%.
- Converted synchronous endpoints into asynchronous background tasks with Celery and Redis queues, enabling autoscaling and eliminating timeout bottlenecks.
- Integrated observability and monitoring using structured logs, metrics, and traces feeding AWS CloudWatch and Grafana dashboards for SLA and incident tracking.
- Improved Kafka-driven workflows with consumer-group batching and backpressure strategies, cutting end-to-end queue latency by ~40%.
- Built lightweight React-based admin tools (feature toggles, health checks) for platform teams to manage internal service states.
- Mentored junior engineers, enforced code quality standards, and drove Agile cross-functional collaboration between backend, frontend, and DevOps teams.
- Built CLI tools (Click) and boto3 scripts to automate service scaffolding, S3 housekeeping, and smoke test runs, reducing onboarding time and manual ops work.

Client: Capital Group

Kafka Platform Engineer

Dec 2023 – June 2024 | Irvine, California

- Architected and launched Stream Hub, a self-service portal (Flask backend + React UI) enabling internal teams to request Kafka topics, register schemas, and auto-generate producer/consumer templates with built-in governance.
- Implemented RBAC and policy-based validations (encryption, retention, ACL templates) to balance guardrails with developer autonomy.
- Delivered Python starter kits using aiokafka and confluent_kafka along with reusable CI templates — reducing integration time from 2 weeks to ~3 days.
- Built “inspect-as-sheet” export to snapshot topic configs, offsets, and ACLs into CSV/Google Sheets for auditing and reporting, cutting manual support tickets by 60%.
- Standardized Kafka Connect (JDBC, S3) pipelines, ensuring observability and consistent consumer-group behavior.
- Automated CI/CD pipelines (Jenkins, GitHub Actions) and container builds; created pre-deploy validation scripts and manifest sanity checkers.
- Authored CLI scaffolding scripts to create topics, ACLs, sample consumers, and CI manifests in one command.
- Conducted onboarding workshops, wrote runbooks, and improved team alignment through API documentation and QA refinements.
- Created utility scripts for batch data validation, JSON response parsing, and nightly job status reporting — accelerating daily verification workflows.

- Built and stabilized data ingestion pipelines that consumed telecom membership/provisioning data (SOAP/XML, flat files) and routed normalized payloads to downstream services under strict SLAs.
- Re-engineered legacy middleware using Python and Flask services to replace brittle Java components; added error resilience, caching, and schema validation, reducing incident rate by ~45%.
- Containerized workloads and deployed on OpenShift, adding liveness probes, graceful restarts, and logging playbooks to improve operational stability.
- Implemented Redis caching and throttling for membership queries, significantly improving throughput under high load.
- Wrote Python-based test harnesses integrating SoapUI + pytest to automate data validation, regression checks, and report generation, saving hours of manual QA.
- Enhanced deployment automation via Jenkins pipelines, Git versioning, and tagging standards, ensuring repeatable, error-free rollouts.

TECHNICAL SKILLS

Languages & Frameworks: Python (FastAPI, Django, Flask, asyncio), Go (Golang), JavaScript (React, Node.js), GraphQL.

Backend & APIs: REST, OpenAPI/Swagger, gRPC, API Gateway patterns, OAuth2 / OpenID Connect, JWT, RBAC, secure API design.

Cloud & Platform: AWS (ECS/Fargate, EKS, Lambda, API Gateway, S3, RDS, IAM, CloudWatch), basic GCP & Azure familiarity.

Containers & Orchestration: Docker, Kubernetes (EKS/GKE), Helm, OpenShift.

CI/CD & IaC: GitHub Actions, Jenkins, ArgoCD, Terraform (modules), CloudFormation / CDK, Helm charts.

Messaging & Streaming: Apache Kafka (topics, schemas, Connect), confluent_kafka, Kafka Streams (awareness), RabbitMQ.

Datastores & Performance: PostgreSQL, MySQL, MongoDB, Redis; indexing, query tuning, caching, connection pooling.

Testing & Quality: pytest, contract testing, unit & integration tests, Postman, SoapUI, JUnit, Mockito, SonarLint / SonarQube.

Observability & Monitoring: Structured logging, CloudWatch, Grafana dashboards, basic tracing (Jaeger/Zipkin), alerting & runbooks.

Automation & Tooling: boto3, Click/argparse (CLI tools), scripting (Python), poetry/pipenv, pandas, PySpark (ad-hoc).

Architecture & Practices: Microservices, async patterns, API versioning, circuit-breaker/retry, canary & feature-flag deployments.

Delivery & Soft Skills: Product ownership, stakeholder communication, mentoring, code review, agile, technical documentation.

EDUCATION

Master of Science in Computer Science | Wichita State University, Kansas, USA | Aug 2021 – May 2023

Bachelor of Technology in Computer Science and Engineering | Amrita University, Tamil Nadu, India | July 2014 – May 2018

CERTIFICATION

- AWS Certified Cloud Practitioner
- Python developer from Infosys
- Oracle Certified Java Developer

PROJECTS & INTERN EXPERIENCE

Graduate Teaching Assistant & IOS Mobile Developer, Wichita State University

Dec 2021 – May 2023 | Wichita, Kansas

- Enhanced iOS app accessibility using Swift, Storyboard, Django enabling features for visually impaired users; supporting Human-Computer Interaction (HCI) research funded by NSF USA.

Java Full Stack Developer, Goldman Sachs

July 2024 – Aug 2024 | Dallas, Texas

- Short-term contract to modernize legacy Java monolith and implement backend integration points for new Python services.
- Implemented a small React admin UI for migration monitoring and feature toggles; ensured backend APIs exposed health metrics used by the admin UI.